

ECON:4800:0001 Fall25 Advanced Causal Inference With Data

INSTRUCTOR

Yuan Liao

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W308

Tippie College of Business

John Pappajohn Business Building

CLASS MEETING TIMES

MW, 3:30-4:45 pm, 60 Schaeffer Hall

DROP-IN HOURS

MW, 10-11:30 am

COURSE SITE

Pearson: Mylab Economics

Course Description and Goals

Econometrics is a set of statistical tools used to analyze economic relationships using economic data. The methods taught in this introductory course can also be employed in the business disciplines of accounting, finance, marketing and management and in many social science disciplines.

Students who successfully complete Econ 322 should be comfortable with basic statistics and probability. They should be able to use a statistical/econometric computer package to estimate an econometric model and be able to report the results of their work in a nontechnical and literate manner. In particular a student who successfully completes Econ 322 will be able to estimate and interpret linear regression models and be able to distinguish between economic and statistical importance. They should be able to critique reported regression

results in applied academic papers and interpret the results for someone who is not trained as an economist.

Textbook/Materials

Introduction to Econometrics

4rd Edition,

- Author: James H. Stock and Mark W. Watson
- Publisher: Pearson
- ISBN 10: 1-292-26445-4

MyLab Economics

The class will be using MyLab Economics for homework assignments. The bundle of the text book and the access to MyEconLab can be purchased online. Instructions of registration to MyEconlab can be found on sakai.

Only **one attempt** will be given to each homework question.

Grading Criteria

Table 1: Grading components

Items	%
weekly assignment	23%
attendance	7%
midterm exam	30%
final exam	40%

Table 2: Grading scale

Items	%
$\geq 90\%$	A
$[80, 90)\%$	B or B+
$[70, 80)\%$	C+ or B
$[60, 70)\%$	C
$< 60\%$	D+ or lower

Course Work

- **Weekly assignment** will be assigned every week on MyLab Economics, and the deadline will be made clearly in each assignment. Homework is often due in a week. Late homework is accepted but 50% penalized.

- Midterm and final exams will be closed book. Students are allowed to formula sheet and a calculator to the exams.
- We will NOT curve.

Lecture Outline

1. Introduction
 - Brief introduction to course; Sources of data.
2. Review of Statistical Concepts
 - Expected value
 - The joint, marginal density, conditional density and independence
 - Covariance and correlation
 - The Normal density
 - Hypothesis tests, confidence intervals
3. The Simple Linear Regression Model
 - The econometric model
 - Estimating the model and interpreting the results
 - Inference and prediction in the Simple Linear Regression Model
4. The General Linear Regression Model
 - Estimating and interpreting the results
 - Interactions among dependent variables
 - Nonlinear effects
5. Assessing Regression Models
 - Omitted variable bias
 - Errors in variables bias
 - Sample selection bias
 - Simultaneous causality bias
6. Additional Topics in Regression
 - Regression Models with Binary Dependent Variable
 - Instrumental Variable estimation
 - Panel data and time series models

How to Get Help

Lecture Notes

I will post lecture notes on Pearson's MyLab Economics every weekend. Going through notes is very important.

Weekly homework

Make sure you understand the questions in homework, especially those questions that you did not get correctly in your first attempt.

Study Plan

Pearson's MyLab Economics also provides practice questions under "study plan". Those are useful for practice.

Tippie Career Services

Tippie Undergraduate Career Services provides students with personalized career support, regularly updated internship and job leads, and other useful career information. Additional information and access to services can be found at <https://students.tippie.uiowa.edu/tippie-resources/career-services/undergraduate-career-services>.

Course-Specific Policies and Guidelines

Weekly assignments: Late homework is accepted but 50% penalized.

Exams:

There will be ONE midterm and ONE final exam.

Students are expected to take the exams at the regularly scheduled times, unless permission has been granted by the instructor at least 2 weeks in advance in writing. All exams are INDIVIDUAL WORK and collaboration on exams is not permitted under any circumstance. All exams are CLOSED BOOK, CLOSED NOTES.

Attendance: Attendance in all class sessions is expected. However, unforeseen conflicts due occasionally arise. Students are expected to give adequate advance notice of an excused absence, make up the missed work and make every effort to avoid a class conflict. To ensure legitimacy, students may be asked to provide documentation upon request. Short-term illnesses will also be considered as excused absences, however students are asked to let me know prior to or even the day of the absence why they are not in class. A plan for managing longer term absences should be discussed with me to determine whether successful completion of the course is possible.

Inclement Weather/Class Cancellation: Although it is our intent to offer every class at its assigned time, on rare occasion there are weather or other emergency events that require that alternative arrangements are made for class delivery. If that happens, please stay alert for an email from me indicating how the class will be handled. This may include: 1) Cancelling the class completely – all reading and work that was due is expected to be completed and turned in on time through ICON; 2); Holding an online version of the class (i.e., via Zoom)– you will receive an email invitation to join the session remotely from a computer with an internet connection – headphones are recommended; 3) using recorded videos and presenting other materials to you through ICON so we can cover the course materials asynchronously.

Collegiate and University Policies and Guidelines

The administrative home of this course is the Tippie College of Business, which governs academic matters relating to the course such as the add/drop deadlines, the second-grade-only option, issues concerning academic misconduct, and how credits are applied for various graduation requirements. Different colleges might have different policies.

The Tippie College of Business is committed to providing students with a diverse, inclusive, and equitable environment in which to pursue their educations. In addition, Students at Tippie adhere to an honor code that emphasizes the importance of honesty and integrity. Student concerns about this class or your performance in it can be discussed with the instructor or the Associate Dean for Undergraduate Programs. More details about these and other policies are available on the Tippie website.

The University of Iowa is committed to the protection of freedom of speech and the principles of academic and artistic freedom, to accommodating students with disabilities, and to accommodating absences due to religious holidays. In addition, students are expected to comply with the University's Code of Student Life. The University is also committed to non-discrimination and prohibits all forms of sexual harassment, sexual misconduct, and related retaliation. The University also provides resources for student mental health as well as for the basic needs and support of students. More information about these and other policies can be found on the Provost's Office's website.